Monopoly: Market Power and Its Consequences

One seller → marginal revenue → output and price → welfare and public policy

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1. Definition of monopoly

The competitive firm in Chapter 15 accepts the market price. A monopolist faces a different problem. It is the only supplier of a good for which buyers have no close substitute. It can influence the selling price by choosing how much to offer, but it cannot force buyers to purchase that output at any price.

> **DEFINITION Monopoly and market power**
>
> A **monopoly** is a market served by **one seller** offering a product with **no close substitutes**. The seller has **market power**: it can influence price. A monopolist is called a **price maker**, in contrast with a competitive **price taker**. This does *not* mean that the monopolist can charge any price it wishes.

Suppose a firm holds the sole enforceable license to supply a specialized industrial corrosion-test cartridge. Buyers can purchase fewer cartridges if the price rises, postpone testing, or change their operations. The seller faces the market's *downward-sloping demand curve*. Raising the price means selling fewer units. Lowering the price permits more sales. Market demand disciplines the monopolist even without a close competing supplier.

> **KEY RESULT The first distinction from Chapter 15**
>
> **Competition:** one small firm faces a horizontal demand line at the market price. **Monopoly:** one firm faces the entire downward-sloping market demand curve. As a result, a monopolist normally must lower the price to sell additional units. That changes its marginal revenue, its chosen quantity, and the price consumers pay.

So what? The objective is still to maximize *economic profit*, not sales or revenue. What changes is the trade-off: selling more requires a lower price, including on units that otherwise would have sold for more.

2. How monopolies arise: barriers to entry

If another firm could immediately supply an identical product, a lone seller's ability to keep its price high would be undermined. Monopoly is sustained by a **barrier to entry**: something that prevents or seriously impedes rival suppliers from entering.

DEFINITION Three sources of monopoly

Control of a key resource: one firm has exclusive access to an essential input or facility that competitors cannot readily duplicate.

Government-created exclusivity: a patent, copyright, or exclusive franchise gives legal protection for certain activities or products. Such protection can create incentives to invent or create, but it also restricts entry.

Natural monopoly: one firm can supply the relevant market at a lower *total cost* than two or more competing firms could. A large fixed infrastructure cost and a low extra cost per customer can create this outcome.

Natural monopoly: why average cost falls

Consider a hypothetical pipeline network carrying treated industrial water to nearby factories. Building the network costs \$120 per period, even before serving its first customer. Supplying one additional customer costs \$3. The cost equations are

$$TC = 120 + 3Q, \qquad ATC = \frac{120}{Q} + 3, \qquad MC = 3.$$

Here Q is the number of equal-sized customer connections. At 30 connections, average cost is $120/30 + 3 = \$7$. At 60 connections, it is $120/60 + 3 = \$5$. The fixed infrastructure cost is being spread over more users. One network serving 60 costs \$300. Two separate networks each serving 30 would cost $2(120 + 3 \times 30) = \$420$.

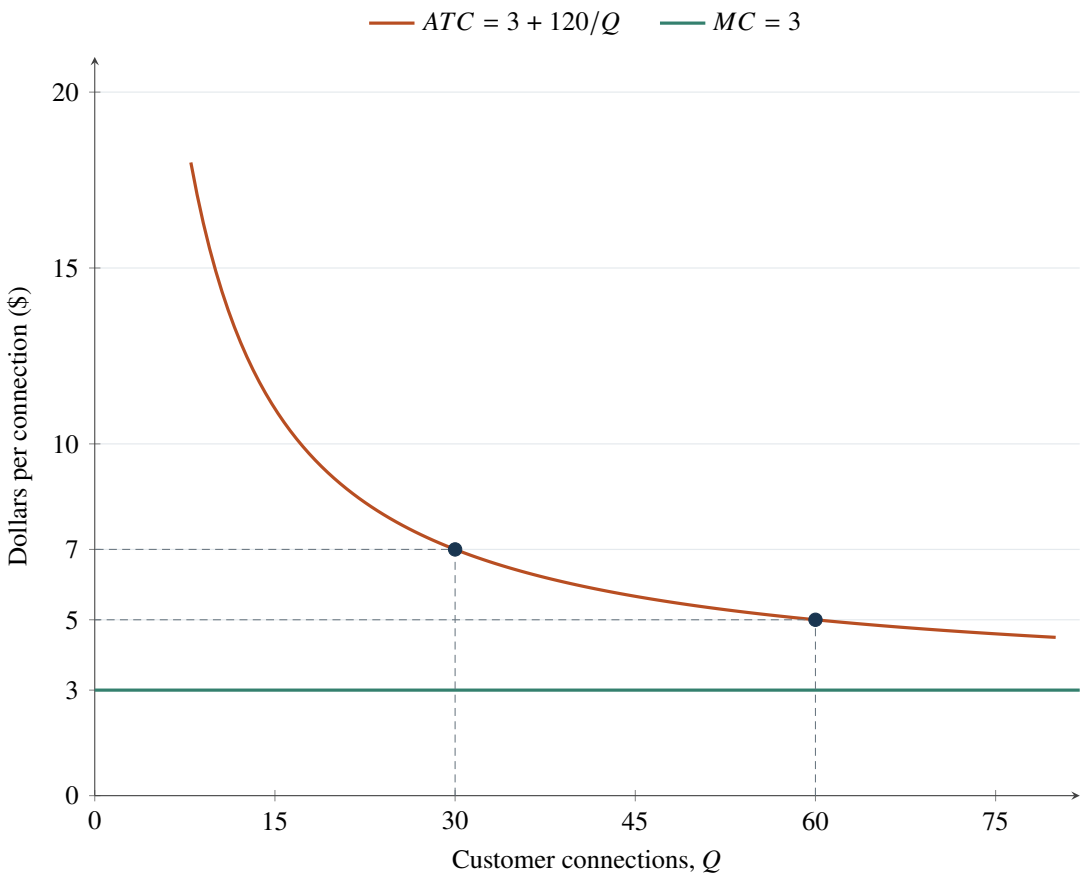


Figure 1. Natural monopoly. As the single network adds customers, fixed infrastructure costs are spread over more connections. Average total cost falls throughout the relevant range.

Read the graph. The orange curve is \$7 at 30 connections and \$5 at 60. The green marginal-cost line stays at \$3. The gap is fixed cost per connection. **So what?** Duplicating a network raises total resource costs when one network can serve the whole market more cheaply. Breaking such a monopoly into competing networks is not automatically efficient.

3. Price versus marginal revenue

The central calculation is about *marginal revenue*. A monopolist can sell another unit only by lowering the price indicated by market demand. Under a single-price policy, that lower price applies to *all* units sold, not just the additional one.

> **DEFINITION Total, average, and marginal revenue**
>
> **Total revenue** is $TR = P \times Q$. **Average revenue** is $AR = TR/Q = P$ for positive output. **Marginal revenue**, MR, is the *change in total revenue* when another unit is sold: $MR = \Delta TR/\Delta Q$. In the tables here, one additional unit means $\Delta Q = 1$.

Worked example: A single-price seller of industrial test cartridges

Use the following original hypothetical demand schedule. The price in dollars for a quantity Q is $P = 32 - 2Q$, and all cartridges must sell at that same price. For example, six cartridges sell at \$20 each, so revenue is \$120. Seven sell at \$18 each, so revenue is \$126. The seventh cartridge adds only \$6 of revenue, not \$18.

Q	Price P	Total revenue TR	Average revenue AR	MR of last unit
0	\$32	\$0	–	–
1	30	30	30	30
2	28	56	28	26
3	26	78	26	22
4	24	96	24	18
5	22	110	22	14
6	20	120	20	10
7	18	126	18	6
8	16	128	16	2
9	14	126	14	−2
10	12	120	12	−6

Where did the \$6 come from? Selling the seventh cartridge brings in \$18 from the additional customer (the *output effect*), but reducing the price from \$20 to \$18 on the previous six loses $6 \times \$2 = \$12$ (the *price effect*). Marginal revenue is therefore $18 - 12 = \$6$. When output goes from eight to nine, revenue falls from \$128 to \$126, so marginal revenue is −\$2. **Marginal revenue can be negative.**

> **KEY RESULT Why the monopoly's marginal revenue lies below its price**
>
> For a smooth downward-sloping demand curve at positive output, $\boxed{MR < P = AR}$. The extra unit raises sales but the price reduction on earlier units lowers revenue. In a discrete table, the first unit is a special case: there are no earlier sales to reduce, so its MR equals its price. Subsequent units have $MR < P$. If the price effect outweighs the output effect, $MR < 0$.

The demand curve and marginal-revenue curve

For diagrams we treat Q as smoothly divisible while keeping the same demand relationship $P = 32 - 2Q$. The smooth marginal-revenue line is $MR = 32 - 4Q$. Students do *not* need calculus: it is a convenient straight line for drawing the changing revenue from additional output. The table reports changes over complete units. For example, moving from five to seven units changes revenue from \$110 to \$126, averaging \$8 of extra revenue per unit around $Q = 6$. This is the smooth MR value at six; the table's sixth- and seventh-unit changes (\$10 and \$6) fall on either side. Thus the smooth line and the whole-unit table agree about the optimal quantity.

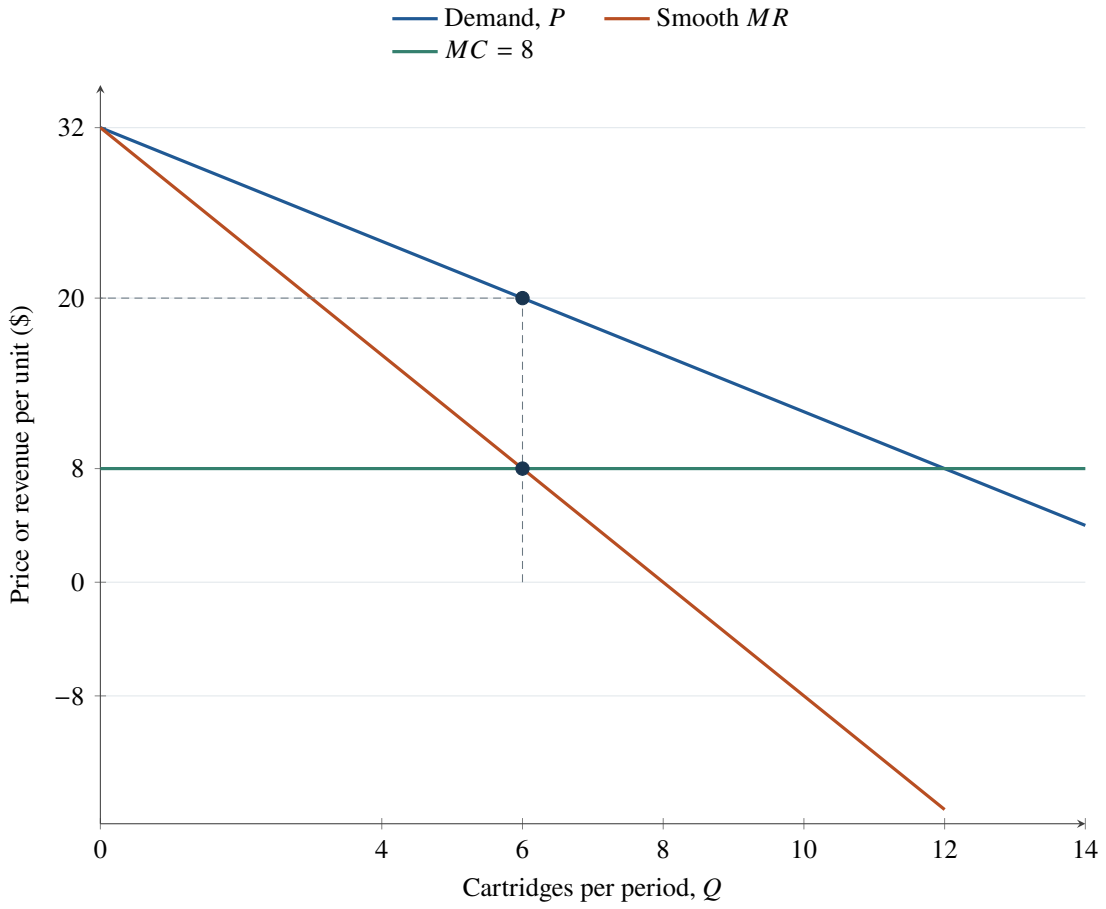


Figure 2. A monopolist's demand is downward sloping, and its marginal revenue lies below demand. At $Q = 6$, the smooth MR line meets the $MC = \$8$ line, but the selling price on demand is \$20.

Read the graph. At $Q = 6$, move up to demand to find the price \$20; move to the orange line to find marginal revenue \$8 in the smooth model. Those are different numbers. On the right, the orange line crosses zero at $Q = 8$; beyond it, selling still more reduces total revenue. **So what?** Drawing price from the MR line would give students the wrong monopoly price.

4. Profit and loss graphically

Monopoly and competition share the *profit-maximizing decision rule*: keep adding output while the marginal revenue from another unit exceeds its marginal cost. The difference is that the monopoly's marginal revenue is below the selling price.

KEY RESULT How a single-price monopolist chooses output and price

Step 1: Choose the quantity where $MR = MC$ on the relevant profit-maximizing part of the curves (or compare next-unit revenues and costs in a table).

Step 2: At that quantity, read the **price from demand**, *not* from marginal revenue.

Step 3: Compare price with average total cost. $\boxed{\pi = (P - ATC)Q}$ gives economic profit or loss.

Worked example: Profit for the cartridge monopolist

Suppose producing each cartridge costs \$8 and, for simplicity, there is no fixed operating cost in this example. Then $MC = ATC = \$8$. At $Q = 6$, the smooth diagram gives $MR = MC = \$8$ and market demand gives $P = \$20$. Profit is $(20 - 8) \times 6 = \$72$. The discrete table gives the same optimizing integer output: the sixth unit adds \$10 in revenue and costs \$8, while the seventh adds only \$6 but costs \$8.

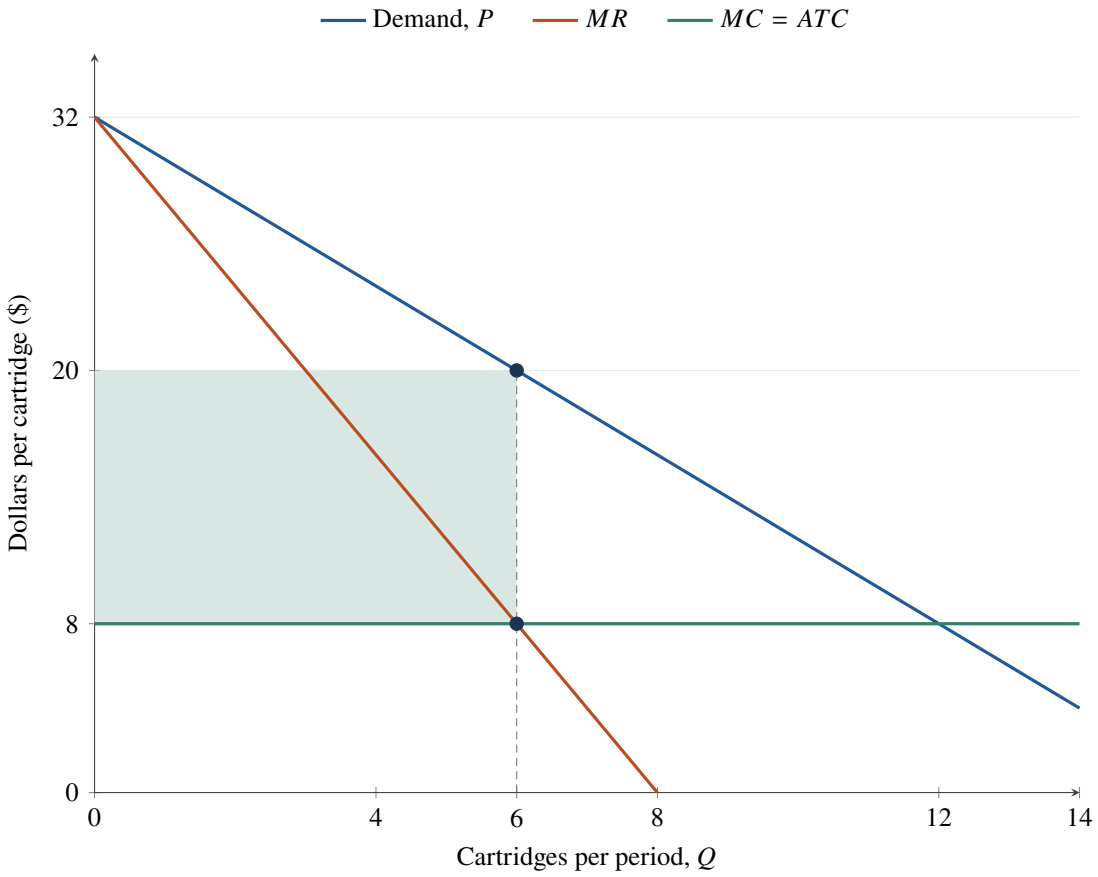


Figure 3. Monopoly economic profit. The height of the shaded rectangle is $P - ATC = \$12$ and its width is six cartridges, giving \$72 profit.

Read the graph. Find $Q = 6$ from the intersection of MR and MC; go up to demand for $P = \$20$. The green horizontal line also gives $ATC = \$8$. The gap of \$12 on each of six cartridges is the profit rectangle. **So what?** The rule $MR = MC$ selects output; the *demand curve* sets what customers pay.

Worked example: A different monopoly can operate at a loss

Consider a separate specialized laboratory with demand $P = 26 - Q$, marginal and average variable cost of \$10 per test, and an unavoidable fixed cost of \$90 per period. Its smooth marginal revenue is $MR = 26 - 2Q$. The profit-maximizing output is $Q = 8$, with price \$18. At that quantity total revenue is $18 \times 8 = \$144$, total cost is $90 + 10 \times 8 = \$170$, and economic profit is $-\$26$. The corresponding average total cost is $170/8 = \$21.25$.

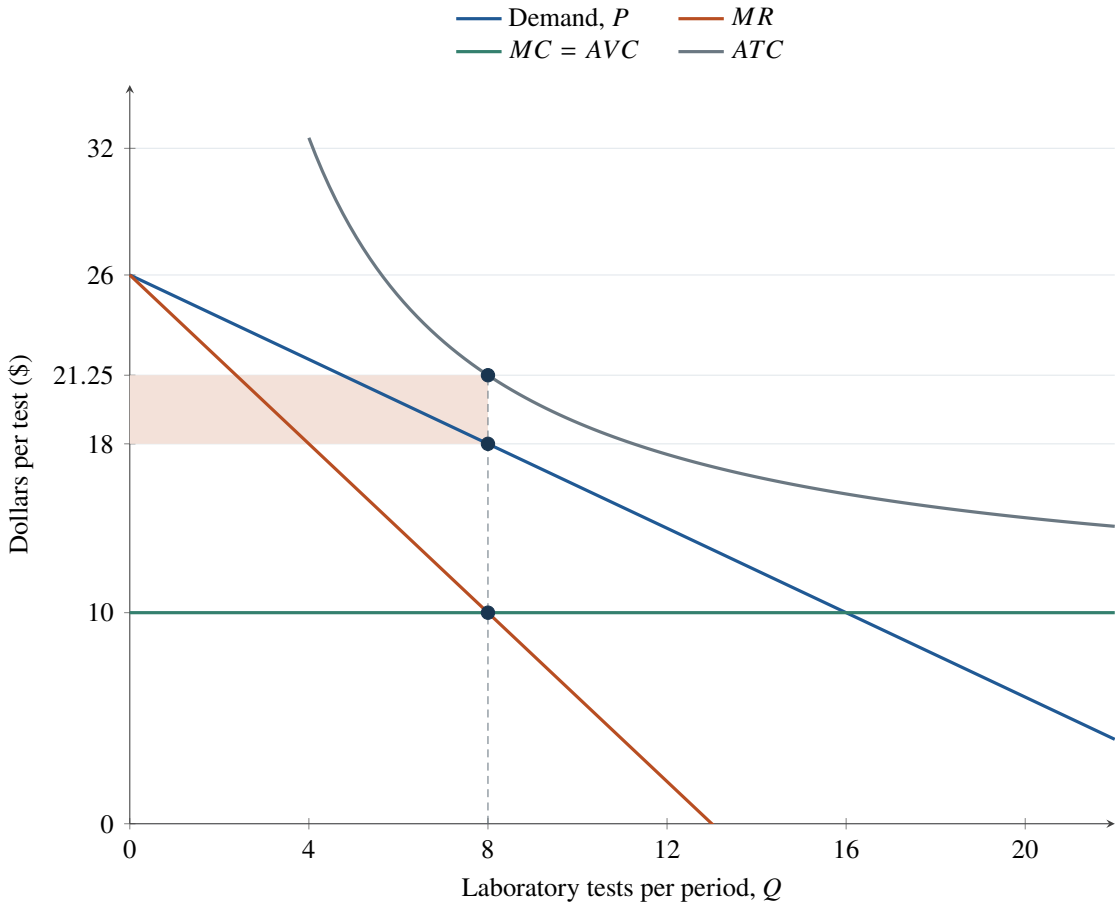


Figure 4. A separate monopolist making a loss. At $Q = 8$, its $P = \$18$ is below $ATC = \$21.25$. The shaded loss rectangle is $(21.25 - 18) \times 8 = \$26$.

Read the graph. The price line on demand lies below the gray ATC curve at the output of eight tests. The firm still operates in the short run: \$18 covers the \$10 variable cost per test. Operating loses \$26, whereas closing temporarily leaves the entire \$90 fixed cost. In the long run, persistent losses induce exit if costs can be avoided.

> **KEY RESULT No monopoly supply curve**
>
> A competitive firm's supply gives a quantity at each *market-given* price. For a monopoly, price and quantity are **jointly determined** by demand, marginal revenue, and marginal cost. Shifting demand can change both and can even yield the same quantity at different prices. Therefore a monopolist does *not* have a supply curve independent of demand.

5. Comparison with perfect competition

A monopolist does not maximize profit by charging the highest imaginable price. It chooses the most profitable combination of price and quantity along market demand. To see how market structure matters, hold demand and marginal production costs fixed and compare the two possible market outcomes.

KEY RESULT The two marginal conditions

Perfect competition: $\boxed{P = MR = MC}$. A price-taking firm sells another unit without lowering its price.

Single-price monopoly: $\boxed{P > MR = MC}$ for an interior optimum on a downward-sloping demand curve. Producing another unit requires a reduction in the price charged on existing sales.

For our original cartridge demand $P = 32 - 2Q$ and marginal cost \$8, monopoly produces six at \$20. In the competitive *marginal-cost benchmark*, price equals \$8 and market quantity is 12. With zero fixed production costs in this particular example, the competitive benchmark also permits zero economic profit. This is a comparison of market outcomes, not a claim that a patent automatically disappears.

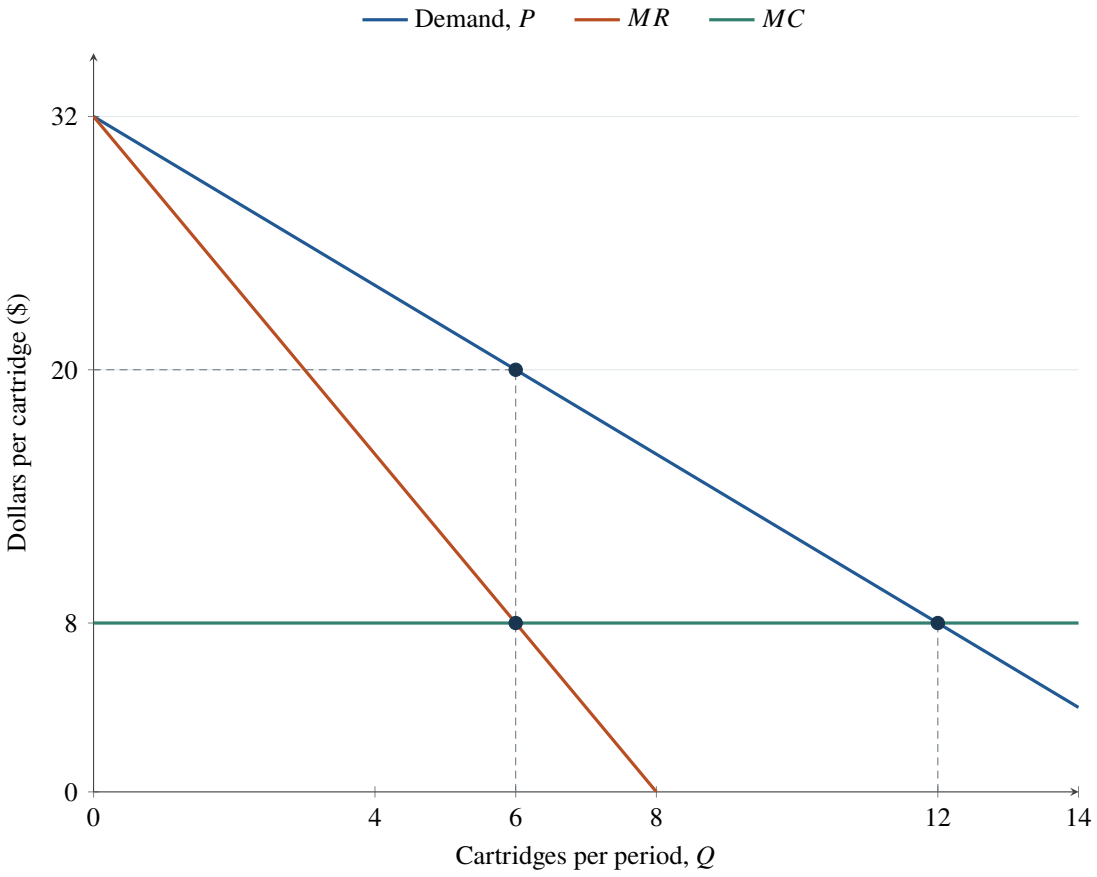


Figure 5. The same demand and marginal cost under two pricing arrangements: single-price monopoly chooses $Q_M = 6$, $P_M = \$20$; the competitive benchmark has $Q_C = 12$, $P_C = \$8$.

Read the graph. Monopoly stops at six where $MR = MC$. Competition reaches 12 where consumers' willingness to pay on demand equals marginal cost. The difference, six forgone cartridges, matters because buyers value those units more than their cost of production.

Feature	Perfect competition	Single-price monopoly
Goal	Maximize economic profit	Maximize economic profit
Output rule	$MR = MC$	$MR = MC$
Can earn short-run profit?	Yes	Yes
Number of sellers	Many	One
Marginal revenue	$MR = P$	$MR < P$
Price relative to MC	$P = MC$ at optimum	$P > MC$ at optimum
Efficient output?	Yes, with standard assumptions	Usually too little
Entry in long run?	Free entry in the model	Blocked by entry barriers
Long-run economic profit	Zero with identical firms, free entry and constant costs	Can remain positive; not guaranteed

So what? Both types of firms want profit and both use marginal reasoning. The difference is market power, which creates a wedge between the consumer's price and the seller's marginal cost.

6. The welfare cost of monopoly

A higher price alone does not describe the full economic problem. Part of a monopoly's additional receipts comes from customers who still buy at the higher price: that is principally a *transfer* between buyers and the seller. The separate efficiency loss comes from trades that never occur.

> **DEFINITION Deadweight loss**
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> **Deadweight loss (DWL)** is the loss of total economic surplus from not producing units whose value to buyers exceeds their marginal cost. In the standard comparison without externalities, the efficient quantity is where **demand meets marginal cost**. A single-price monopolist normally produces less.

The cartridge market has $Q_M = 6$, $P_M = \$20$, $Q_C = 12$, and $MC = \$8$. Between six and 12 cartridges, willingness to pay declines from \$20 to \$8. Producing those units would create additional gains from trade. The triangle has base $12 - 6 = 6$ and height $20 - 8 = 12$, so its area is $\frac{1}{2} \times 6 \times 12 = \$36$.

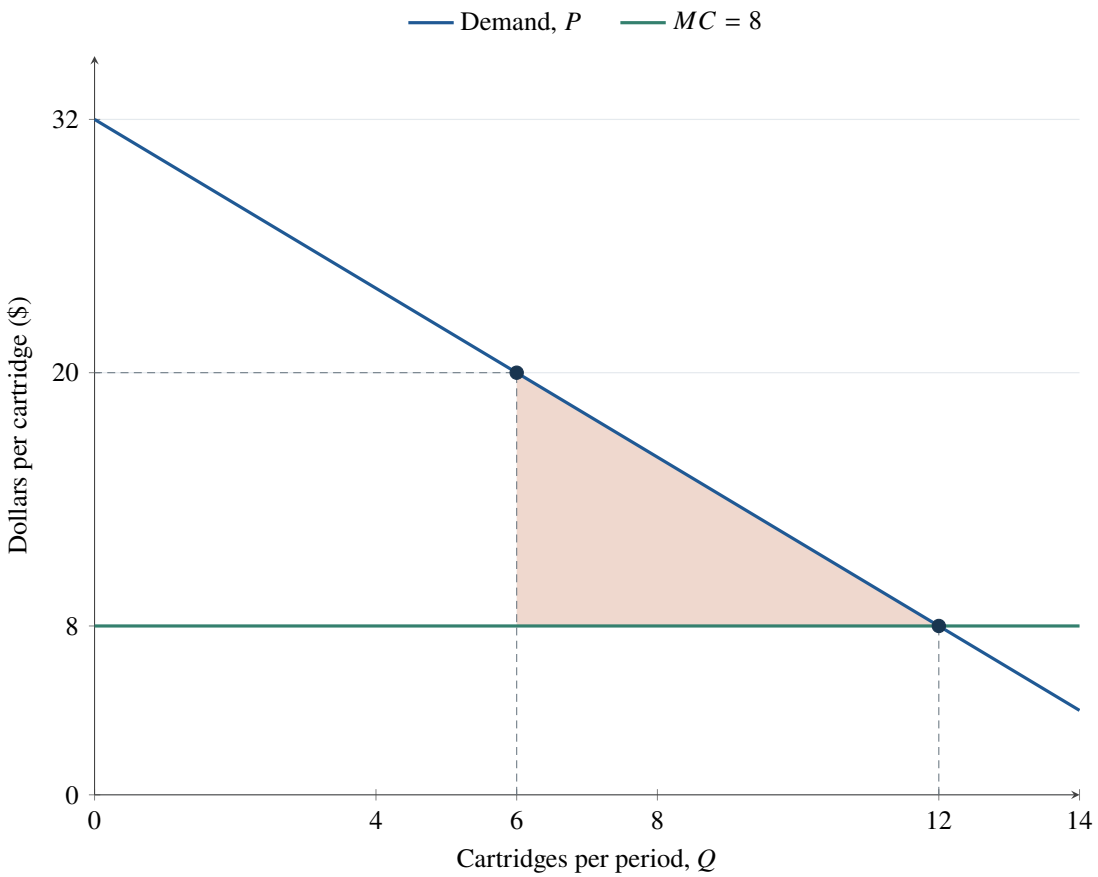


Figure 6. The shaded deadweight-loss triangle is $\frac{1}{2}(12-6)(20-8) = \$36$. The six units between monopoly output and the efficient quantity would have been worth more to buyers than they cost to make.

Read the graph. Over the shaded range, the demand curve lies above the cost line. Those are unrealized gains from trade. The monopoly's \$72 profit in the original example is *not* the \$36 deadweight loss. Higher seller profit may include transfers from consumers; only forgone net gains reduce total surplus in this simple model.

KEY RESULT Distinguish distribution from efficiency

Who receives surplus? Higher monopoly prices transfer some surplus from consumers to the seller.

How large is total surplus? Output restriction creates deadweight loss because some customers who value additional units above their cost cannot buy them. A transfer alone does not destroy social surplus; resources spent acquiring or defending monopoly power can create additional real costs.

7. Price discrimination

A single-price monopolist charges every customer the same price for identical units. Yet customers often differ in willingness to pay. If the firm can separate them, different prices can sometimes increase profit and allow additional purchases.

DEFINITION Price discrimination

Price discrimination means charging different prices for the same good when the difference is *not explained by differences in marginal cost*. It requires some market power and a way to separate buyers or transactions according to willingness to pay. Effective separation usually also requires preventing low-price buyers from reselling to high-price buyers.

Worked example: Two customer groups for an industrial testing service

Suppose three factories will each buy one audit at up to \$70, while four smaller facilities will each buy one at up to \$30. Each audit costs \$10 to deliver; assume no fixed cost. A single price of \$70 sells three audits and makes $(70 - 10) \times 3 = \$180$ profit. A single price of \$30 sells seven and makes $(30 - 10) \times 7 = \$140$. If the service can reliably charge the first group \$70 and the second \$30, with no resale, profit becomes $3(70-10)+4(30-10) = \$260$. Different prices increase profit by \$80 relative to the best uniform price, and four extra audits are completed.

KEY RESULT Price discrimination can change both profit and output

Charging a higher price to customers with greater willingness to pay and a lower price to other customers can raise the seller's profit. It can also serve buyers excluded by a single high price. Its effect on overall economic welfare is **not always positive**; it depends on how quantities change, not merely on how payments are redistributed.

Perfect price discrimination: a useful limiting case

Under **perfect price discrimination**, the seller knows each unit's willingness to pay and charges exactly that amount. It sells units as long as the willingness to pay exceeds marginal cost. For the original cartridge model, output therefore rises to $Q = 12$, where demand meets $MC = \$8$. Consumer surplus is zero, because buyers pay their entire willingness to pay; the seller captures the total gains from trade. There is no deadweight loss in this idealized case, assuming no additional costs of sorting and charging buyers.

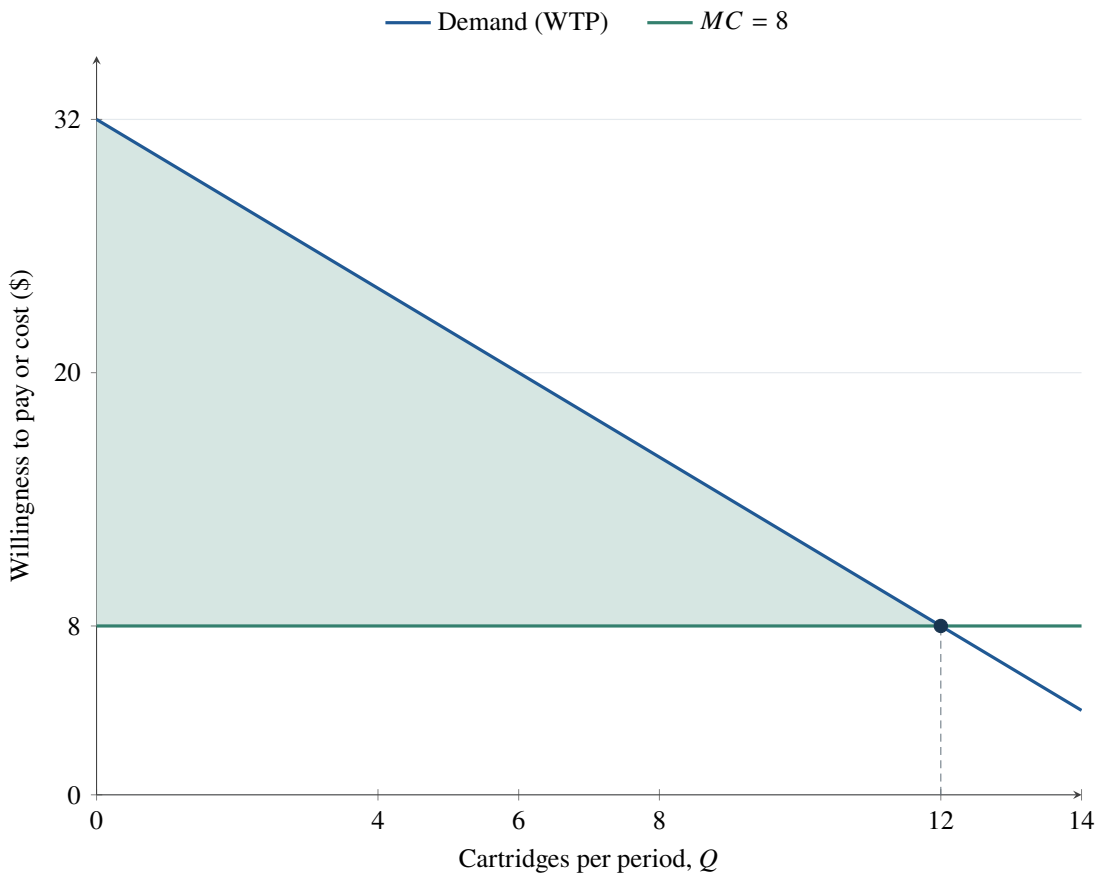


Figure 7. Perfect price discrimination in the same cartridge market. Every unit worth at least marginal cost is sold, giving $Q = 12$ and no deadweight loss. The shaded area is $\frac{1}{2} \times 12 \times 24 = \$144$ of surplus captured by the seller.

Read the graph. The highest willingness to pay is \$32 and marginal cost is \$8. The shaded triangle is the total value above production cost for 12 cartridges. With perfect discrimination, the firm receives that amount rather than leaving consumer surplus. **So what?** A seller can capture more surplus *and* produce the efficient quantity. Distribution and efficiency are separate questions.

In practice, firms do not know each customer's precise willingness to pay. More realistic forms include negotiated contracts for business equipment, different refundable and nonrefundable transport fares, and volume-based pricing where the per-unit price changes with quantity. Different service costs must be distinguished from genuine price discrimination.

8. Public policy toward monopolies

The appropriate policy depends on why monopoly exists. A protected seller of a product that several firms could supply presents a different problem from a utility where duplicating pipes or cables would increase costs.

DEFINITION Four policy approaches

Antitrust: prevent anticompetitive conduct or mergers and, in suitable cases, restore competitive alternatives.

Price regulation: limit the price of a monopoly, particularly a natural monopoly, while recognizing the firm's need to cover its costs.

Public ownership: have a public authority operate the service, with accountability and management incentives replacing private-profit incentives.

No intervention: leave the outcome unchanged where intervention's likely costs and unintended effects exceed its expected benefits.

Why marginal-cost pricing can make a utility lose money

Return to the original pipeline with $TC = 120 + 3Q$, $MC = \$3$, and $ATC = 3 + 120/Q$. Suppose a regulator requires the firm to serve 60 connections at a price equal to marginal cost, \$3. Revenue is $60 \times 3 = \$180$, but total cost is $120 + 3 \times 60 = \$300$. The utility loses \$120. At this output, average total cost is \$5, so charging only \$3 cannot finance its fixed network.

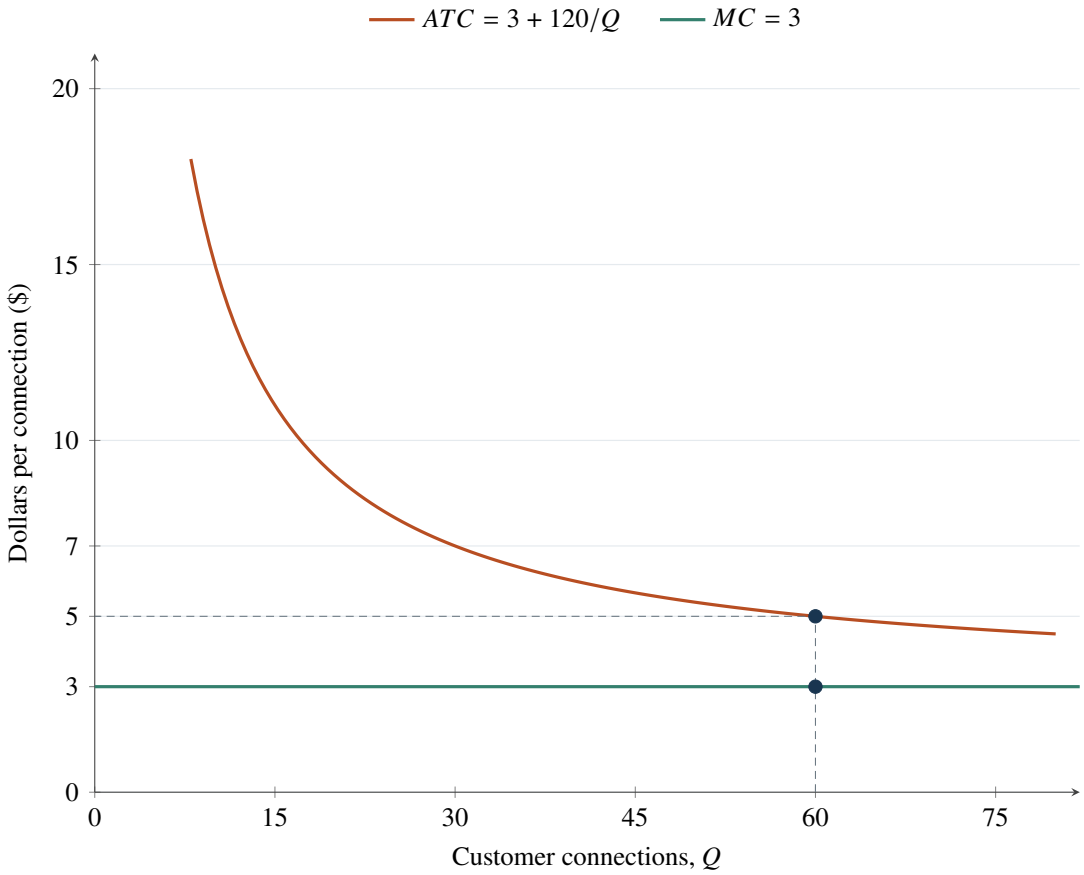


Figure 8. The same natural-monopoly cost curves as Figure 1. At 60 connections, marginal cost pricing at \$3 is below average total cost of \$5. The gap of \$2 per connection implies a \$120 deficit.

Read the graph. The green line gives the cost of one more connection. The orange curve adds the average cost of

the existing network. When price equals the green line, the utility cannot recover the $120 fixed cost. This is the economic reason marginal-cost regulation may require a subsidy.

> **KEY RESULT The natural-monopoly policy trade-off**
>
> **Marginal-cost pricing** encourages efficient use but may require a tax-funded subsidy to cover fixed costs. **Average-cost pricing** can cover costs without a subsidy but normally sets price above marginal cost, leaving some worthwhile demand unserved. Other options include fixed access fees plus usage charges, where feasible. No method eliminates the trade-off between efficient marginal use and financing fixed infrastructure.

So what? Antitrust may help when entry is unnecessarily restricted; forced duplication may waste resources in a natural monopoly. Public ownership does not make resources free and regulation does not remove the underlying cost structure. Whether intervention improves welfare depends on its actual costs and incentives.

Quick review: identify the decision

Question	Rule and economic reason
Which output maximizes profit?	Compare added MR with MC; for a smooth interior solution use $MR = MC$.
Where is monopoly price on a graph?	Read it from the *demand curve* at the chosen quantity.
How is profit or loss measured?	$(P - ATC)Q$, using ATC at that same quantity.
Does monopoly have a supply curve?	No price-to-quantity relation independent of demand.
Where is deadweight loss?	Between monopoly and efficient output, where demand lies above MC.
What does perfect discrimination do?	Under ideal assumptions, it expands output to the efficient quantity and captures buyer surplus.
Why can marginal-cost regulation lose money?	A natural monopoly may have $MC < ATC$ because fixed infrastructure costs must be recovered.

Three common mistakes, corrected

Mistake 1: "Monopoly price equals MC." No. $MR = MC$ selects output; demand gives the price. In our example: $P = \$20$, $MC = \$8$.

Mistake 2: "Monopoly profit equals deadweight loss." No. Much profit is transferred from consumers; DWL comes from forgone trades. Here profit is $72 and DWL is $36.

Mistake 3: "Split every natural monopoly." No. Duplicating a network may raise total cost. Regulation must also finance shared infrastructure.

> **KEY RESULT One final connection**
>
> Both firms maximize profit using $MR = MC$. In competition, $P = MR$; in a single-price monopoly, $P > MR$. Monopoly usually restricts output below the efficient quantity, creating deadweight loss. The entry barrier helps determine whether intervention can improve the outcome.